CSE-538-01-4265 Graph Databases and Analytics (3 credits)
Summer 2026 (Tue & Thu, 4 pm – 5:50 pm, Sackett 103)

Texts

[1] Class PowerPoints (posted on the Blackboard)
[2] TOMAŽ BRATANICˇ and OSKAR HANE, *Essential GraphRAG*, Manning 2025 (https://neo4j.com/essential-graphrag/ for a free e-copy)

Neo4j Manuals and Online Courses

[N1] Neo4j Cypher Manual v5 (https://neo4j.com/docs/pdf/neo4j-cypher-manual-5.pdf)
Neo4j Cypher Manual v25 (https://neo4j.com/docs/pdf/neo4j-cypher-manual-25.pdf)
[N2] APOC User Guide 2026.03 (https://neo4j.com/docs/pdf/neo4j-apoc-2026.03.pdf)
[N3] The Neo4j Graph Data Science Library Manual v2026.03 (https://neo4j.com/docs/graph-data-science/current/)
[N4] Neo4j Python Driver Manual 6 (https://neo4j.com/docs/pdf/neo4j-driver-manual-6-python.pdf)
[N5] Free Neo4j Courses (https://graphacademy.neo4j.com/categories/)

References

[R1] Ian Robinson, Jim Webber & Emil Eifrem, *Graph Databases, second edition*. O'Reilly Media, Inc., 2015
(follow this link https://neo4j.com/lp/book-graph-databases/ to get a free PDF copy)
[R2] Alessandro Negro, *Graph-Powered Machine Learning*, Published by Manning Publications Co. 2021.
[R3] Building Knowledge Graphs: A Practitioner's Guide By Jesús Barrasa & Jim Webber Publisher: O'Reilly, 2023 (https://neo4j.com/knowledge-graphs-practitioners-guide/)
[R4] Demystifying Graph Databases: Analysis and Taxonomy of Data Organization, System Designs, and Graph Queries (https://arxiv.org/pdf/1910.09017.pdf)

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Office Hours

Tue and Thu: 1 pm - 3pm or by appointments

Software:

- Neo4j Desktop 2.14 (https://neo4j.com/deployment-center/), or
  Neo4j Desktop 1.6.3 (https://neo4j.com/deployment-center/)
- Neo4j Python clients v6.1

Course Prerequisite: Graduate standing or CSE 302

Course Description:

Our world is connected. Data representing real-world problems for analysis, however, usually are discrete and do not explicitly include connected information (relationship or link) in their data models. Graph analytics is the study and analysis of data that can be transformed into a graph representation consisting of nodes (to represent real-world entities) and edges (to represent relationships between entities). Graph analytics is built on the mathematics of graph theory, with augmentation of properties attached to nodes and edges.

This course covers three main topics, which form the foundation of graph analytics. These three topics are listed and described below:

Topic 1 Graph theory and graph platforms and processing

In this topic, we overview graph theory and graph processing platforms and processing. In particular, we introduce Neo4j and Python graph libraries and their appropriateness for different graph processing requirements.

Topic 2 Graph data models and graph database system – neo4j

In this topic, we cover how to store graph data in large storage devices. In particular, we will learn the Neo4j graph database system (its graph data model, graph database engine, and cypher graph data language extended with graph algorithms library functions) to model real-world problems.

Topic 3 Graph algorithms, knowledge graphs, and AI applications

In this topic we study graph algorithms (particularly how they are applied to data stored in neo4j databases), knowledge graphs, and their applications to real-world problems. The applications studied may include E-commerce real-time recommendation systems, pattern detection knowledge graphs (fraud detection), dependency knowledge graphs, Neo4j and LLM (Large Language Model), Embedding and Vector Database, Retrieval-Augmented Generation (RAG), etc.

Grading Policy (The grade will be possibly with +/-.)

90% and above	A
80%-89%	B
70%-79%	C

60%-69%	D
Below 60%	F

The grading is based on the following assignments scale:

A) Class projects	70%
B) One term project	30%

Class Logistical Plan

1. Teaching Method – 100% in-person or 100% on-line depending on which sessions you take.
2. Technology requirements for Distance ED. sessions: Microsoft Teams
3. The Blackboard Announcement and e-mail will be used for communication between students and the instructor.

Computer Issues and IT Support

Speed IT staff are available by appointment from 9 am to 4 pm to assist you with your technology needs. You may schedule an appointment by sending a detailed email including any relevant error codes and screen snips at SPDHelp@Louisville.edu (preferred) or 502-852-7620.

Title IX/Clery Act Notification

Sexual misconduct (including sexual harassment, sexual assault, and any other nonconsensual behavior of a sexual nature) and sex discrimination violate University policies. Students experiencing such behavior may obtain confidential support from the PEACC Program (852-2663), Counseling Center (852-6585), and Campus Health Services (852-6479). To report sexual misconduct or sex discrimination, contact the Dean of Students (852-5787) or University of Louisville Police (852-6111).

Disclosure to University faculty or instructors of sexual misconduct, domestic violence, dating violence, or sex discrimination occurring on campus, in a University-sponsored program, or involving a campus visitor or University student or employee (whether current or former) is not confidential under Title IX. Faculty and instructors must forward such reports, including names and circumstances, to the University's Title IX officer.

Topics by week (This schedule may be changed as needed.)

Week (Monday)	Topic (Tue and Thu classes)
1. May 11	Introduction: Graph Analytics; Graph theory
2. May 18	Using Neo4j desktop and graph Data Model
3. May 25	Neo4j cypher and neo4j python client
4. June 1	Neo4j cypher and neo4j python client
5. June 8	Neo4j cypher and neo4j python client
6. June 15	Graph Data Science
7. June 22	Graph Data Science
8. June 29	Embedding/Vector Database/RAG-AI applications
9. July 6	Embedding/Vector Database/RAG_AI applications
10. July 13	Embedding/Vector Database/RAG-AI applications
11. July 21	Last day of class: July 20 (Mon); Final exam: July 21 – 25 (Tue – Sat)